

⚡ AI-POWERED CYCLONE RESILIENCE PLATFORM

PRAHAR

Predictive Risk & Anticipatory Hazard Action Resource

Track-Based Cyclone Impact & Infrastructure Vulnerability Forecaster for Bay of Bengal & Coastal APAC

🌐 Google Earth Engine 30m

⚡ Gemini 1.5 Flash Multimodal

🌊 Hydrodynamic Bathymetry

🇮🇳 22 Indic Languages Audio

Designed for District Magistrates, NDRF Commanders & State Disaster Authorities (OSDMA, WBDMD, APSDMA, TNDMA) to trigger anticipatory evacuations < T-12h before landfall.

Core Objectives



1. Ingest Track Telemetry

Real-time ingestion of IMD bulletins, JTWC advisories, tide gauges, and Copernicus 30m GEE DEM elevation models.



2. Surge Modeling

Hydrodynamic coastal inundation envelopes across 3 surge hazard tiers (>2.5m, 1.5–2.5m, 0.5–1.5m).



3. Infrastructure Graph

Pinpoint severed CHCs, tripped substations, inundated roads, and cutoff cyclone shelters before surge strikes.



4. Multimodal SOPs

Gemini 1.5 Flash auto-drafts actionable tactical directives for NDRF, DMs, and Panchayats in 22 Indic tongues.



5. Parametric Escrow

Automated pre-landfall liquidity escrow releasing emergency disaster cash at T-24h with zero claims latency.



Our Mission: Empower State & District Disaster Authorities with track-based predictive intelligence, multimodal AI, and transparent parametric liquidity — turning raw forecast data into life-saving anticipatory action.

PROBLEM STATEMENT

The Coastal Vulnerability Gap

● The Problem Today

▲ Fragmented Telemetry Silos

Cyclone tracks from IMD, bathymetry from INCOIS, power grids, and hospital directories exist in isolated databases without unified spatial correlation.

▲ 12–24h Bureaucratic Response Lag

Standard disaster dashboards update post-facto. Ground damage assessments occur 24–48 hours after coastal embankments are breached.

▲ Static Paper–Based Emergency SOPs

District Collectors rely on generic paper manuals unsuited for live dynamic water levels, leading to trapped civilians and flooded evacuation routes.



"The storm surge inundated the coastal arterial highway before the evacuation buses were dispatched."

Without predictive, track-based infrastructure vulnerability modeling, relief forces deploy blindly after lifelines sever. Coastal districts lose critical power and emergency medical access within hours.

The PRAHAR Solution

1. Unified Geoprobe

Direct ingestion of live IMD track coordinates, JTWC advisories, Copernicus 30m DEM elevation, and OpenStreetMap infrastructure networks into a unified spatial graph.

2. Multimodal AI Triage

Gemini 1.5 Flash evaluates hydrological choke points, drafting role-directed SOPs and generating multi-lingual Web Speech broadcasts across 22 official Indian languages.

3. Parametric Liquidity

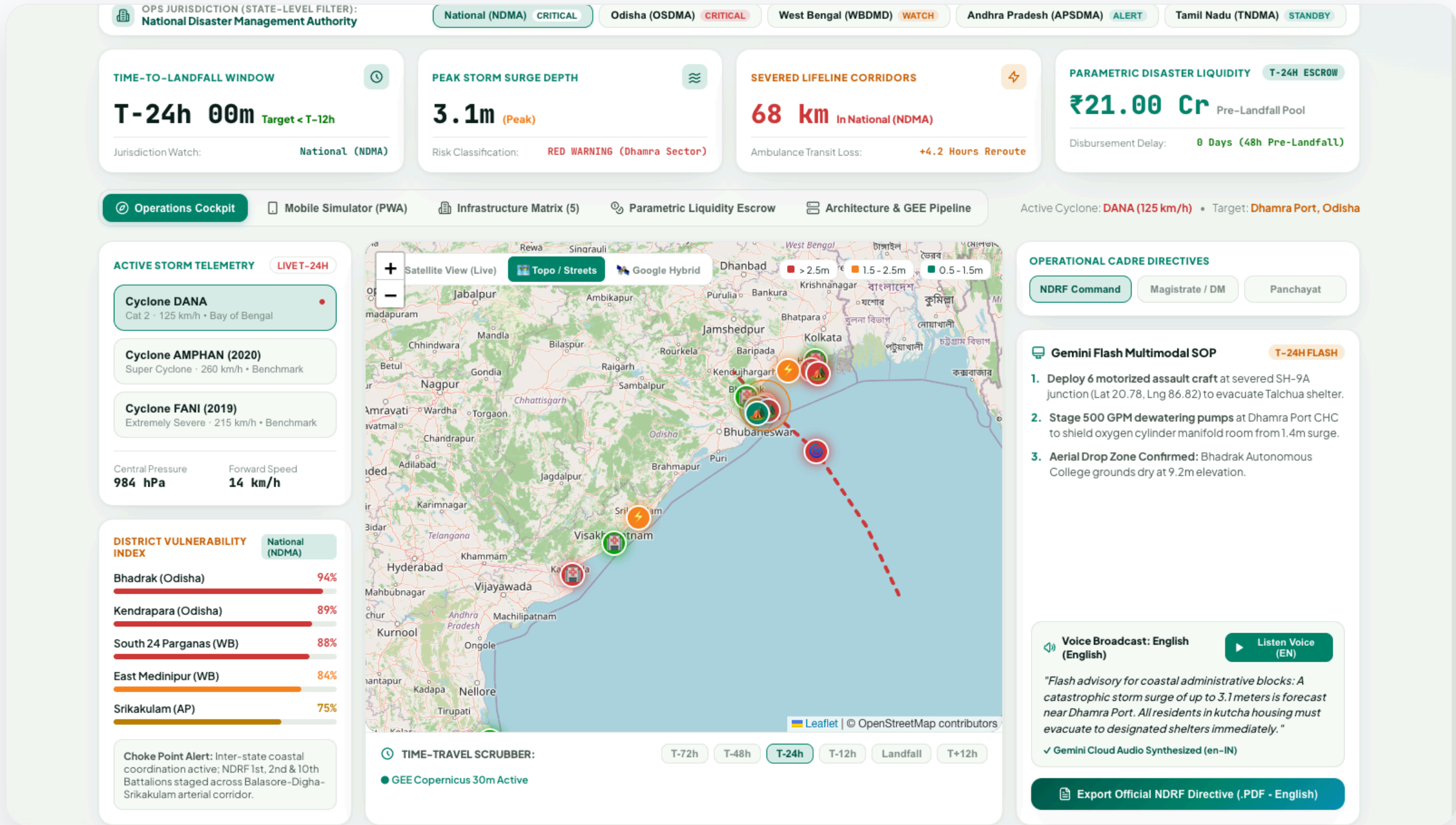
Automated pre-landfall escrow releases emergency cash reserves at T-24h directly to SDRF accounts without bureaucratic verification delays or post-facto claims.



The 5–Minute District Collector Value: You don't have to decipher raw radar files. PRAHAR instantly highlights which bridge is severed, which hospital loses backup power, and where amphibious assault craft must be pre-staged.

INTERACTIVE SHOWCASE

Live Operations Cockpit



Real-Time Capabilities

• Dynamic Storm Tracking

Live trajectory of Cyclone DANA (125 km/h) with 6-stage temporal scrubber from T-72h to Landfall.

• 3-Tier Surge Envelopes

Catastrophic (>2.5m red), Severe (1.5–2.5m orange), and Saline (0.5–1.5m green) inundation polygons.

• District Vulnerability Index

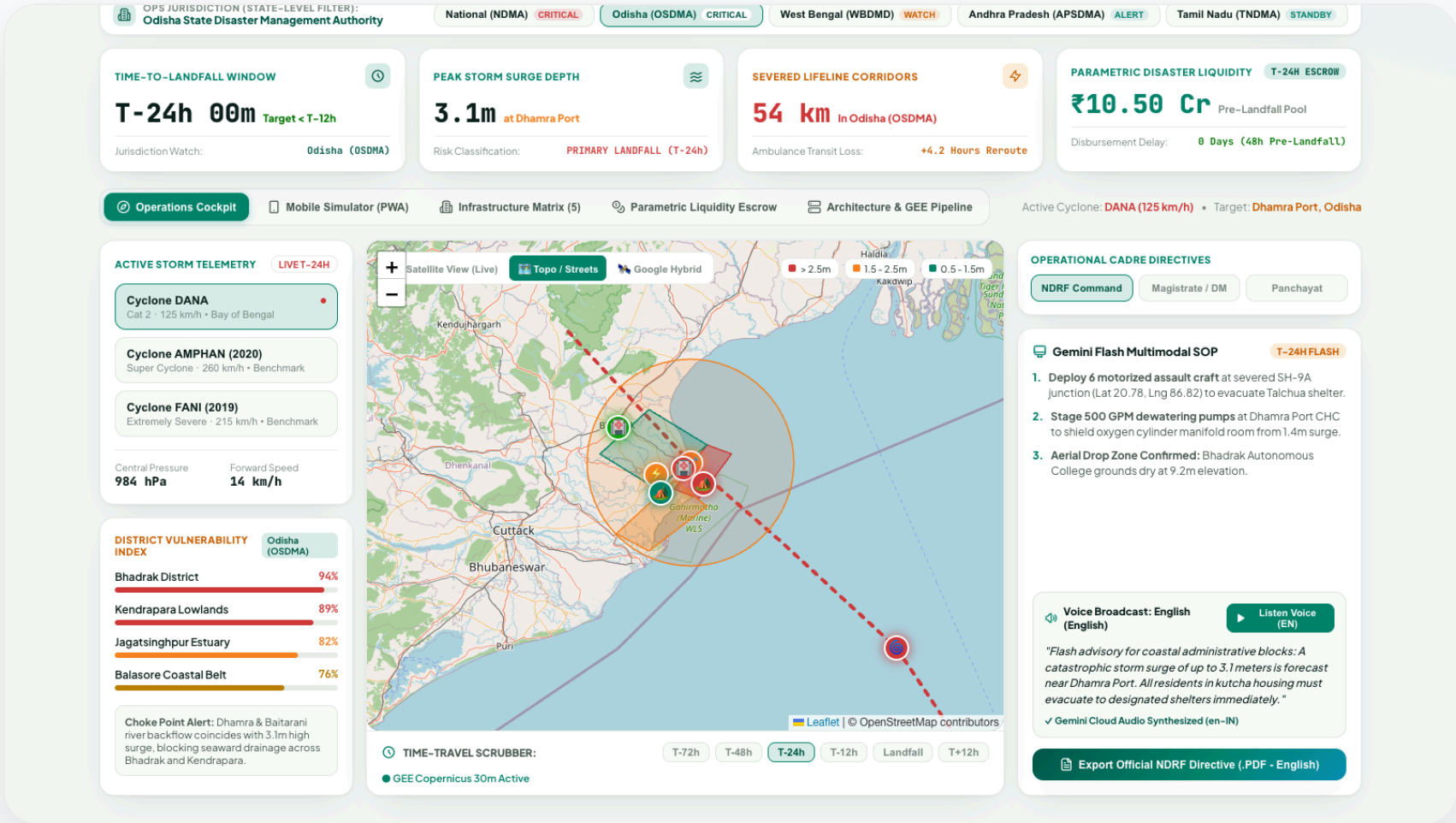
Ranks Bhadrak (94%), Kendrapara (89%), and South 24 Parganas with hydrological choke point alerts.

• Multimodal SOP Directives

Role-based tactical instructions for NDRF, District Collectors, and Panchayats.

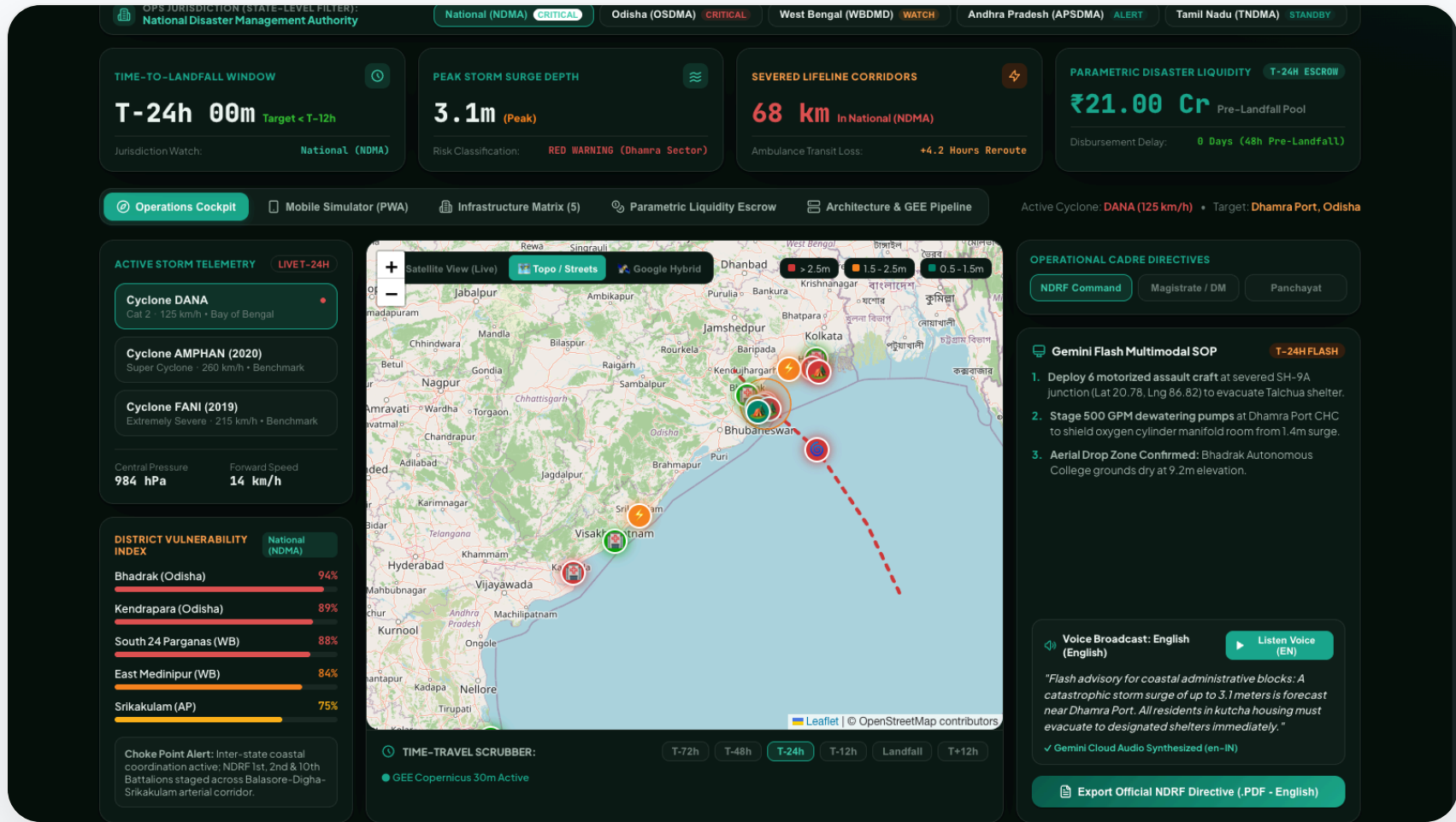
Integrated with Leaflet GIS & OpenStreetMap Topo View

State Operations & 22 Official Indic Languages



Multi-State Operational Jurisdictions

1-click filtering across National (NDMA), Odisha (OSDMA), West Bengal (WBDMD), Andhra Pradesh (APSDMA), and Tamil Nadu (TNDMA) automatically re-centers GIS and updates state-specific choke points and asset markers.

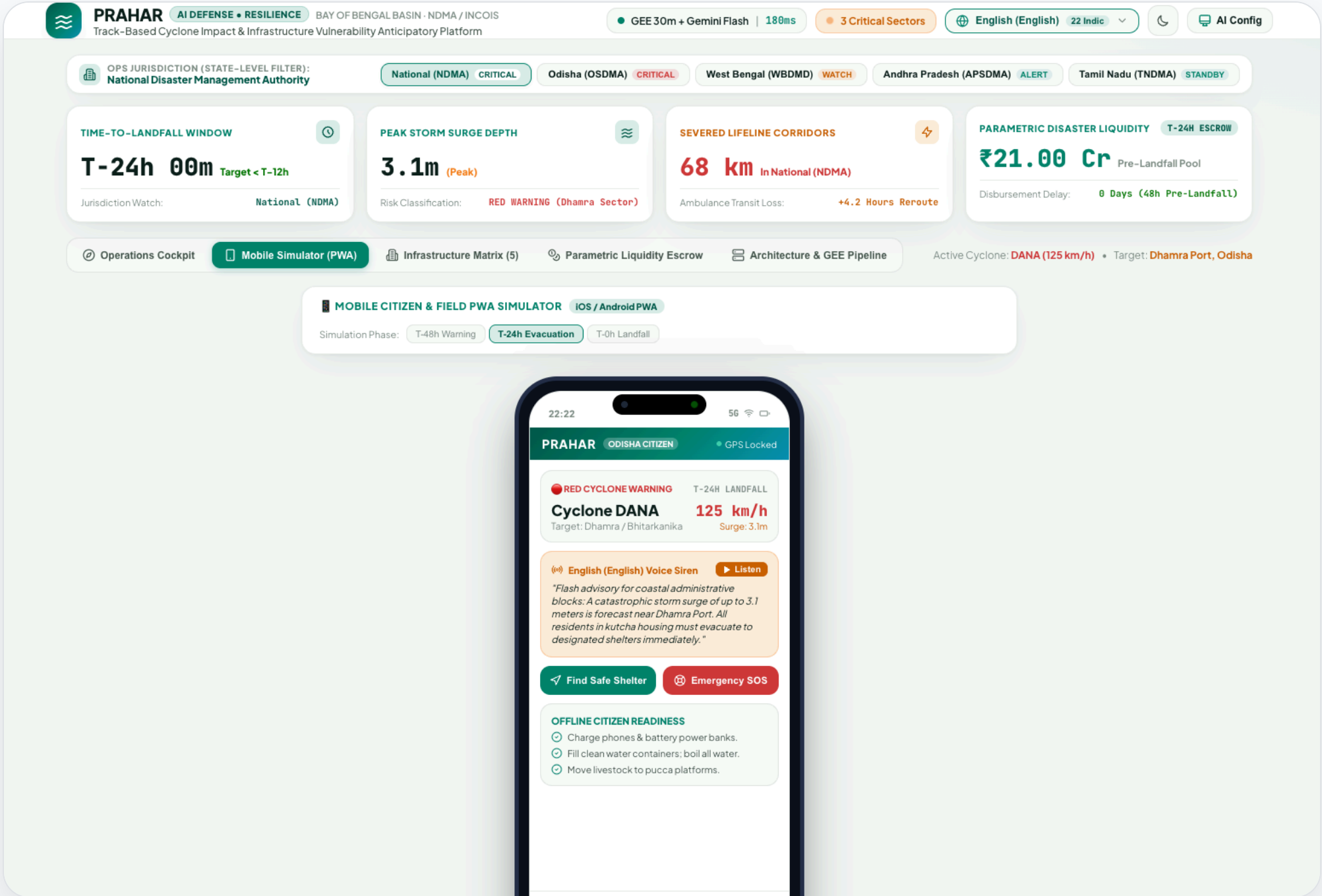


All 22 Eighth Schedule Indian Languages

Native translations and instant Web Speech audio broadcasts for Odia, Bengali, Telugu, Tamil, Hindi, Marathi, Gujarati, etc. Dark Mode supports 24/7 command center screening without visual fatigue.

CITIZEN LAST-MILE

Mobile Citizen Simulator (PWA)



📱 Citizen Protection Features

✓ Offline-First PWA Architecture

Pre-caches elevation maps and shelter waypoints. Fully functional even when mobile towers lose power or backhaul connections.

✓ Dynamic Shelter Navigation

Real-time routing diverts citizens away from flooded roads toward elevated shelters with verified remaining capacity.

✓ One-Tap SOS Distress Beacon

Broadcasts GPS coordinates, battery level, and family count to district emergency operations centers.

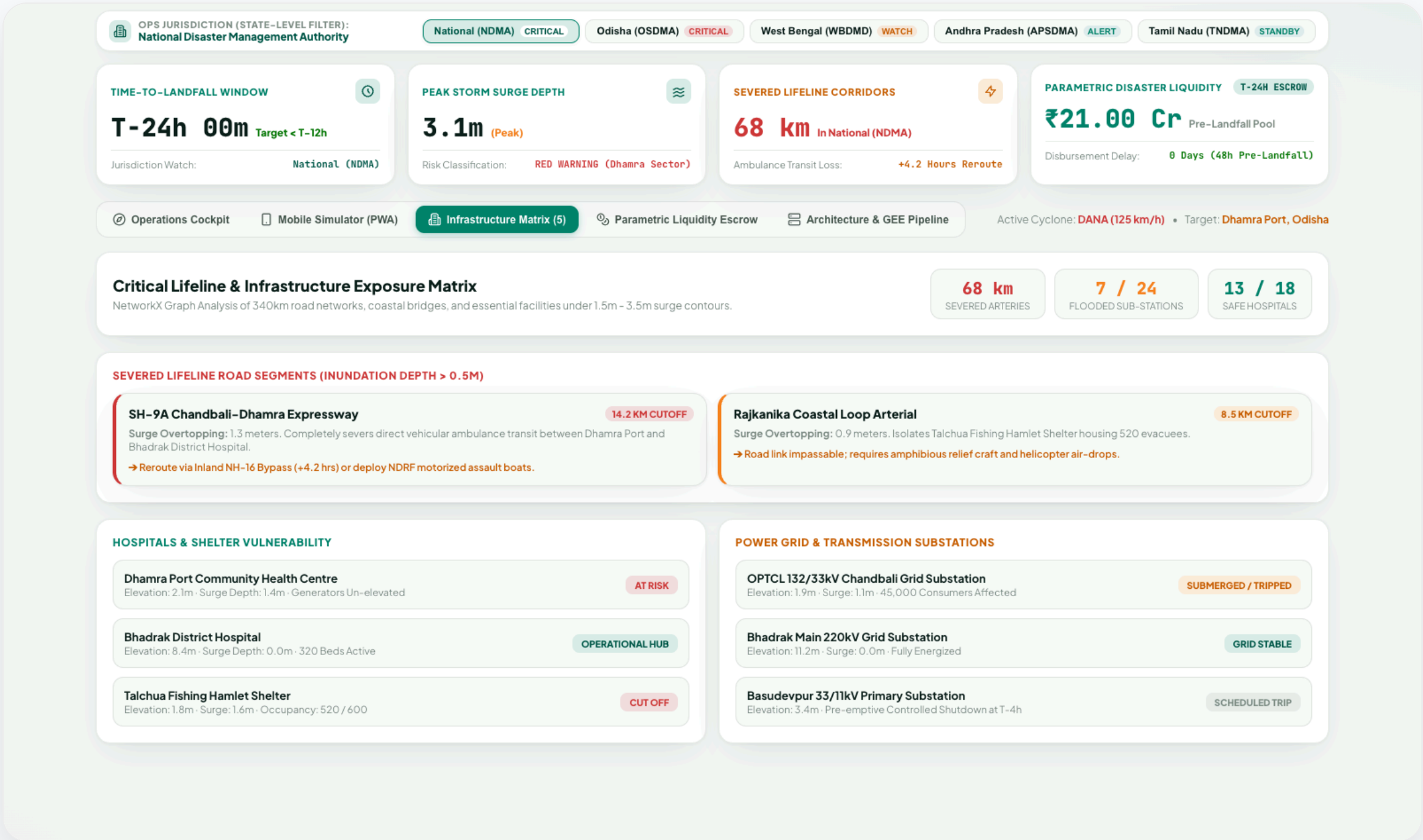
✓ Parametric Relief QR Pass

Digital QR credentials enable immediate ration and relief fund verification at local evacuation shelters.

Simulated on iPhone 16 Chassis with Dynamic Island Telemetry

EXPOSURE ASSESSMENT

Infrastructure Vulnerability Matrix



5 Critical Lifeline Sectors

Primary Health Centers (CHCs)

Elevation vs surge calculation flags oxygen manifold flooding and power cutoffs before breach.

33/11kV Substation Grids

Tripping predictions prevent equipment burnout and identify affected consumer counts.

Multi-Purpose Cyclone Shelters

Live capacity tracking and road severance warnings dispatch amphibious assault craft.

Arterial Highways & Bridges

Identifies specific road cuts (e.g. SH-9A) causing ambulance reroutes of 4.2+ hours.

Physics-Based Inundation Thresholding (DEM vs Storm Surge)

Tab 3: Infrastructure Matrix (?tab=infrastructure)

System Architecture & Decoupled Pipelines

TIER 1: TELEMETRY

External Feeds

- **IMD / JTWC:** Real-time cyclone tracks & cone coordinates
- **GEE API:** Copernicus DEM 30m, Sentinel-1 SAR, GPM Rain
- **OSM Overpass:** Hospitals, substations, arterial bridges
- **INCOIS / NOAA:** Real-time coastal tide station bathymetry

TIER 2: INGESTION (:8001)

Ingestion Microservice

- **Async Worker:** Dedicated background poller engine
- **Track Normalizer:** Cones & waypoints parser
- **Raster Pipeline:** GEE elevation & bathymetry sampler
- **Feature Extractor:** Lifeline critical assets extractor

TIER 3: CORE SIMULATION (:8010)

FastAPI Core Engine

- **SLOSH Surge Model:** Coastal inundation simulator
- **Compound Flood:** River-surge backwater engine
- **NetworkX Graph:** Road severance & isolation solver
- **Gemini Flash:** Multimodal advisory & SOP generation

TIER 4: PRESENTATION (:5174)

React 18 + TS + Vite

- **National War Room:** State-level ops command cockpit
- **Leaflet GIS Map:** 3D inundation & severance layer
- **Temporal Scrubber:** T-72h to Landfall simulation
- **22 Indic Audio:** Web Speech multimodal broadcasting

⚡ Full Data Flow:

IMD / GEE / OSM Feeds → Ingestion Worker (:8001) → PostGIS & Redis Cache → FastAPI Engine (:8010) → Gemini AI SOPs → War Room UI (:5174)

End-to-End System Data Flow Diagram

